Technology Stack

Date	28 June 2026
Team ID	SWTID-2026-2260
Project Name	Smart Lender
Maximum Marks	2 Marks

Define Your Technology Stack:

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	HTML5, CSS3, Bootstrap,	*Provides a responsive, user-friendly interface for applicants and bank officials to enter loan details and view prediction results.*

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	*Python, Flask*	Flask handles HTTP requests, integrates the trained machine learning model, processes applicant data, and returns loan approval predictions efficiently.
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	**SQLite** *(or MySQL if your project uses MySQL)*	*Stores applicant details, prediction results, and loan history securely while supporting efficient data retrieval.*
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	IBM Cloud	Hosts the Smart Lender application, enabling scalable, reliable, and cloud-based deployment for real-time loan prediction.

5	**Version Control & CI/CD**	Git, GitHub	*Supports source code management, team collaboration, version tracking, and simplifies project maintenance*
6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	*Scikit-learn, XGBoost, Pandas, NumPy, Joblib*	*Scikit-learn and XGBoost are used to build and evaluate ML models. Pandas and NumPy handle data preprocessing and analysis. Joblib is used to save and load the trained XGBoost model for deployment.*